

Quant Roadmap Solutions

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Note: There are definitely better ways to do some of these problems. I wrote them out in a complicated manner, sometimes reiterating basic/obvious truths, to make sure I didn't trick myself into understanding something that I actually didn't.

1 Brainteasers

Tigers & Sheep: 100 tigers and 1 sheep are placed on a barren island. If a tiger eats the sheep, it transforms into a sheep. All tigers are perfectly rational and aim to: (1) stay alive, and (2) eat the sheep if it's safe to do so. Given these conditions, will the sheep be eaten?

Answer:

A tiger will only eat the sheep if it can guarantee that it, itself, will not be eaten. So, a tiger will only decide to eat the sheep in this case if when there are 99 tigers, no tiger decides to eat the sheep. And a tiger will only choose to eat the sheep when there are 99 tigers if, when there are 98, no tiger chooses to eat the sheep, and so on. So, we must reason inductively upwards from the number of tigers; let that be n . When $n = 1$, it's safe to eat the sheep since no other tigers exist. When $n = 2$, it's not safe as if a tiger eats, the other one will eat it. When $n = 3$, it is safe, because if a tiger eats then it reduces to the $n = 2$ case. So, for any n that is safe, the $n + 1$ th iteration is not safe as we add a last tiger to finish off the additional sheep. Conversely, for any n that isn't safe, the $n + 1$ th must be safe because we add another tiger to remove the tiger finishing off the sheep (and thus it will never finish it off). This means we alternate between safe and unsafe, and since $n = 1$ is safe, all odd numbers are safe for the tigers (i.e. sheep gets eaten) and all even numbers aren't safe. Consequently, when $n = 100$, the sheep will not get eaten.

Hopping Rabbit: A rabbit sits at the bottom of a staircase with n stairs. The rabbit can hop up only one or two stairs at a time. How many different ways are there for the rabbit to ascend to the top of the stairs?

Answer:

The key to solving in problem lies in thinking inductively. Since we can only take one or two steps at a time, to get to the n step, we can either start from

the $n - 1$ th stair and take a single step, or start from the $n - 2$ th stair and take a double step (observe that we could also have taken two single-steps, but after we take the first of these steps, we're actually at the $n - 1$ th step, which double counts the total number of ways). So, there is one way to get from the $n - 1$ th step to the n th step, and total number of ways to get to the n th step is the same as the number of ways to get to the $n - 2$ th step plus the number of ways to get to the $n - 1$ th step. This is true inductively. So, if we let $f(n)$ be the number of ways to climb n steps, then $f(n) = f(n - 1) + f(n - 2)$, which is the same recurrence as the Fibonacci sequence!

Pirate Loot: 5 pirates loot a chest full of 100 gold coins. They agree on the following method to divide the loot: The oldest pirate will propose a distribution of coins. Then, all of the pirates will vote on whether or not to accept his proposal (the oldest pirate votes as well). If at least 50% of the pirates vote in favor, the proposal will pass. If not, the oldest pirate is killed and the process starts over with the next oldest pirate. All pirates are rational: they want to stay alive with highest priority, get as much gold as possible with second highest priority, and have as few pirates on the boat as possible with third highest priority. How will the coins be divided in the end?

Answer:

To solve this, we must think inductively. First, let's consider the youngest pirate. If they are the only one remaining, they can simply take all of the coins. The same is the case if the two youngest pirates remain, except this time the second-oldest pirate takes all of the gold (they can simply propose this and vote for themselves as they just need a 50% approval rate to win). So, their proposal is guaranteed. Now, the third-youngest pirate knows that the youngest pirate will guaranteed make no money (but will still be alive) if they allow the second-youngest pirate to propose. So, if he gives just one coin to the youngest pirate and keeps the rest for himself, he guarantees said pirate is on his side.

Now, consider the fourth-oldest pirate. They know that if they're decision isn't passed through, then the second-oldest pirate will get nothing. So, they can simply keep 99 coins and give one to the second-youngest pirate to guarantee approval.

Now, the fifth oldest pirate just has to offer two coins - to the third oldest and youngest pirate, because if they allow decision to go to the second-oldest pirate they will be left with nothing, and as such, since they are rational, their only choice is to vote. So, in the end, the oldest pirate keeps 98 coins, one goes to the third oldest, and one goes to the youngest.